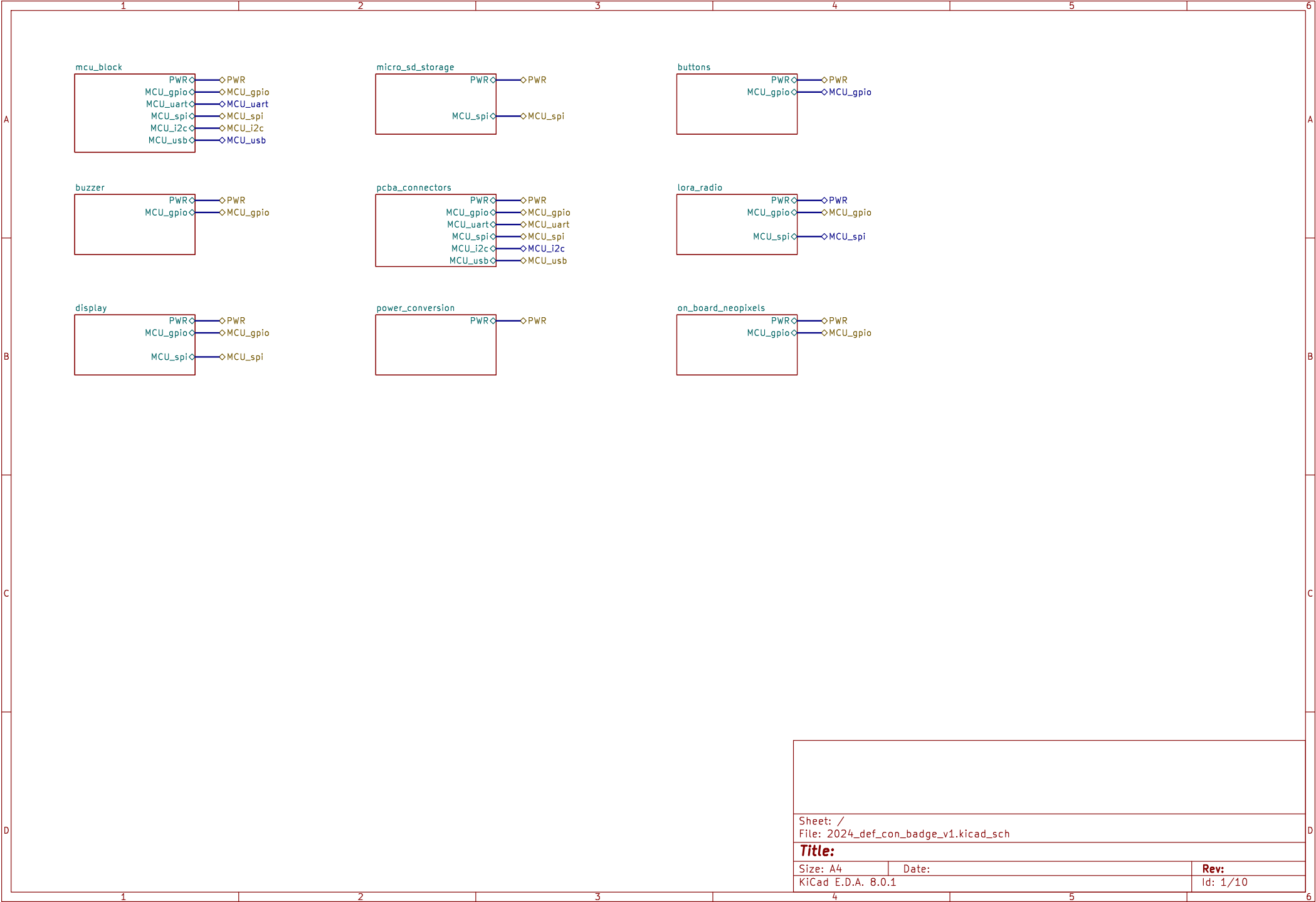
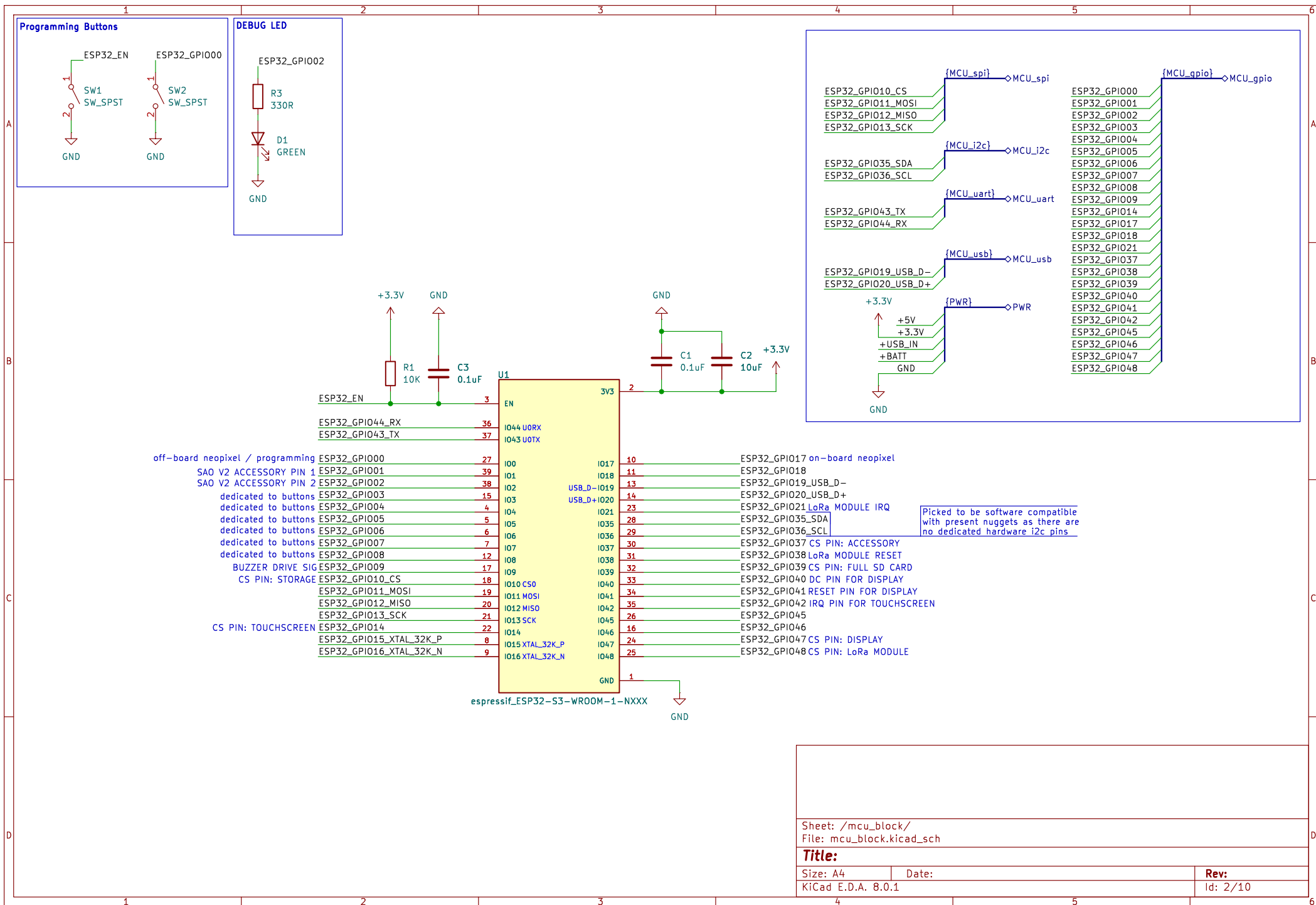
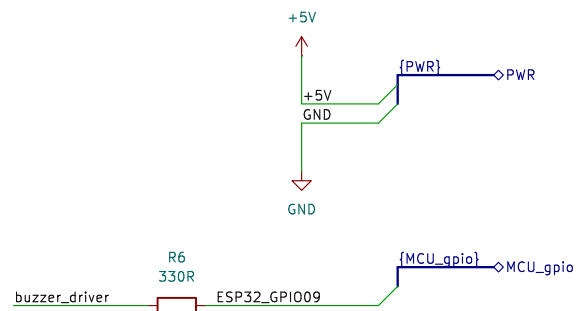
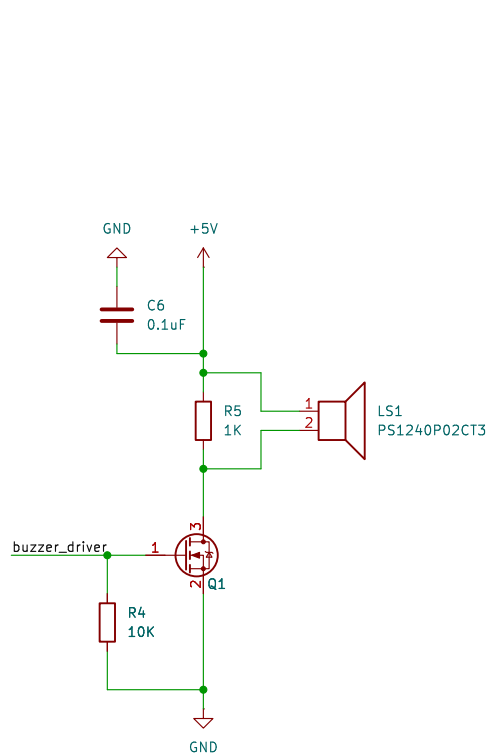








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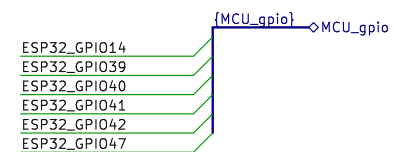
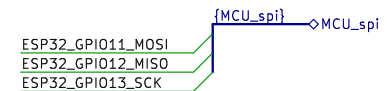
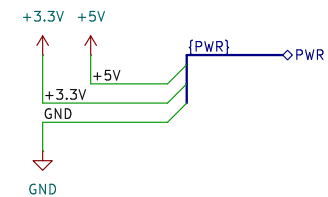
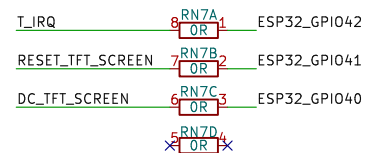
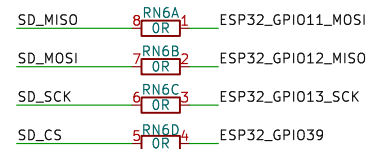
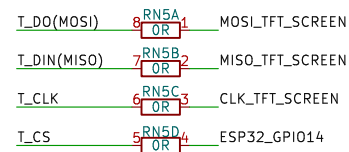
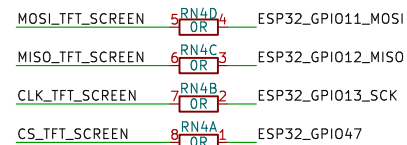
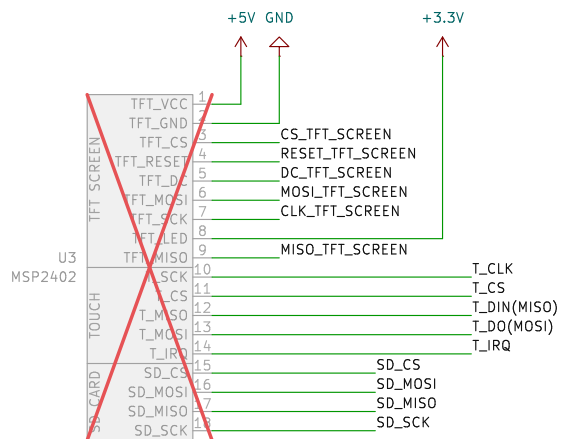
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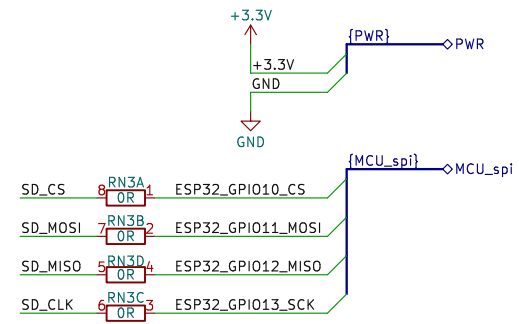
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The diagram shows the wiring for a USB to UART bridge using an ESP32 module. The ESP32 is connected to a USB module (J7) via the following connections:

- VBUS (A4):** Connected to +USB_IN.
- CC1 (A5):** Connected to the D+ pin of the USB module.
- CC2 (B5):** Connected to the D- pin of the USB module.
- D- (A7):** Connected to ESP32_GPIO19_USB_D-.
- D+ (A6):** Connected to ESP32_GPIO20_USB_D+.
- D- (B6):** Connected to ESP32_GPIO20_USB_D+.
- SBUS1 (A8):** Connected to GND.
- SBUS2 (B8):** Connected to GND.

The USB module (J7) is also connected to a USB shield (SHIELD) and a USB cable. The shield is connected to GND (A1) and the cable is connected to GND (S1).

AA (2x) BATTERY INPUT

aa_batt_pos

Resettable Fuse

SW9

F1 Polyfuse

+BATT

D3 RED

D4 GREEN

R24 330R

R2 330R

Q3

Reverse polarity indicator

1

2

3

GND

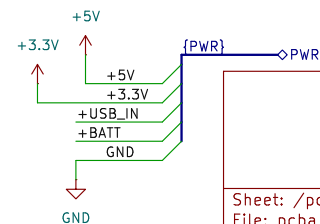
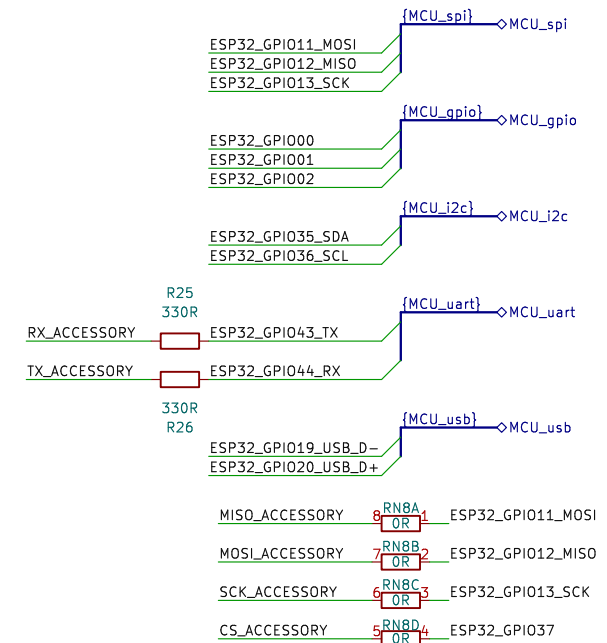
J9

1

2

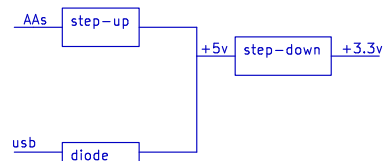
aa_batt_gnd

AA (2x)

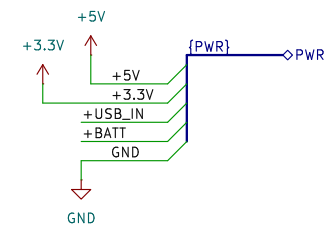
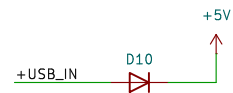


Id: 6/10

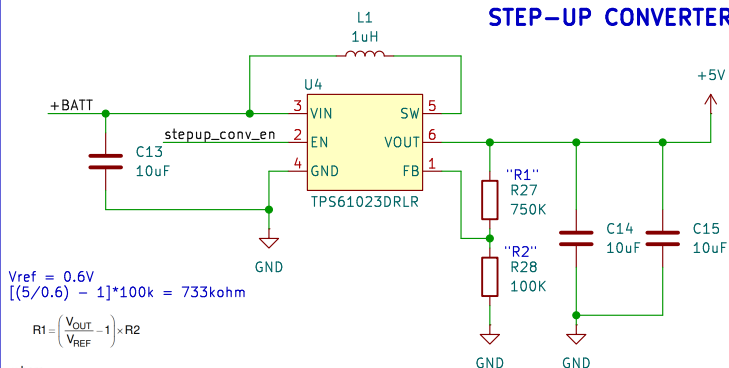
-----General concept for power layout-----
Diode prevents fighting between usb and dc/dc converter
step-up converter only on when usb power is disconnected



USB POWER



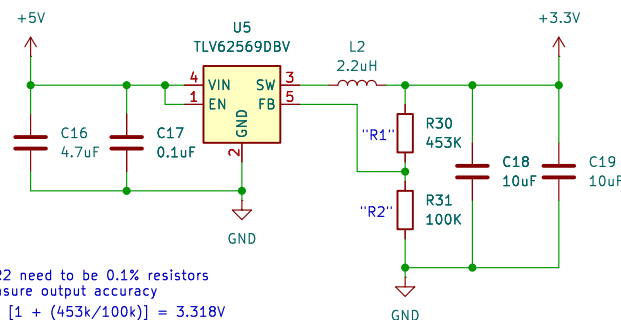
STEP-UP CONVERTER



- V_{OUT} is the regulated output voltage
- V_{REF} is the internal reference voltage at the FB pin

Set to 5.1V to ensure biasing
of diode with usb power

STEP-DOWN CONVERTER

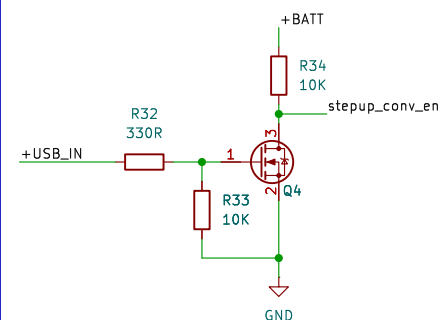


R1/R2 need to be 0.1% resistors
to ensure output accuracy

$$0.6 \cdot [1 + (453k/100k)] = 3.318V$$

$$V_{OUT} = V_{FB} \times \left(1 + \frac{R1}{R2}\right) = 0.6V \times \left(1 + \frac{R1}{R2}\right)$$

STEP-UP CONVERTER ENABLE



Sheet: /power_conversion/
File: power_conversion.kicad_sch

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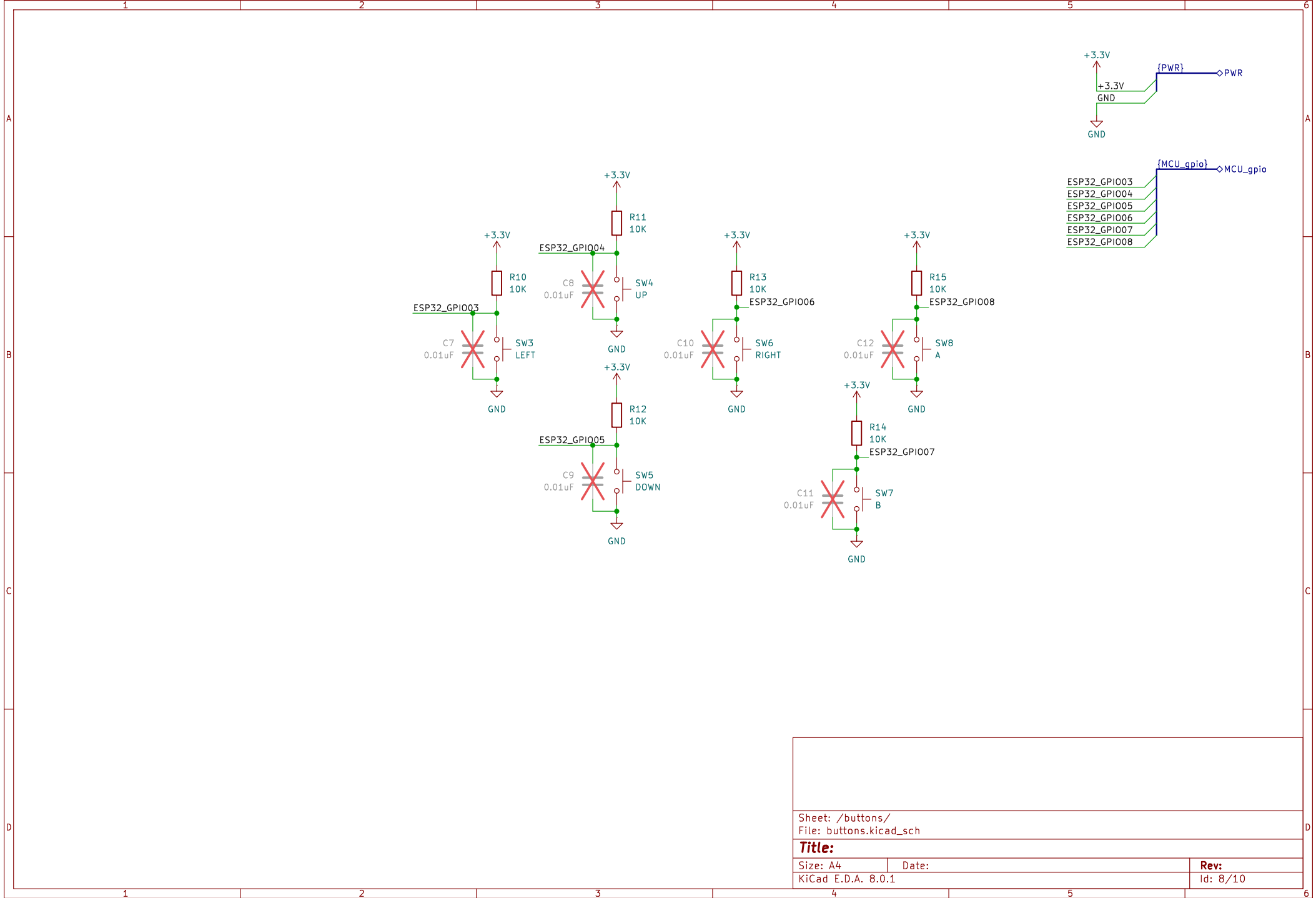
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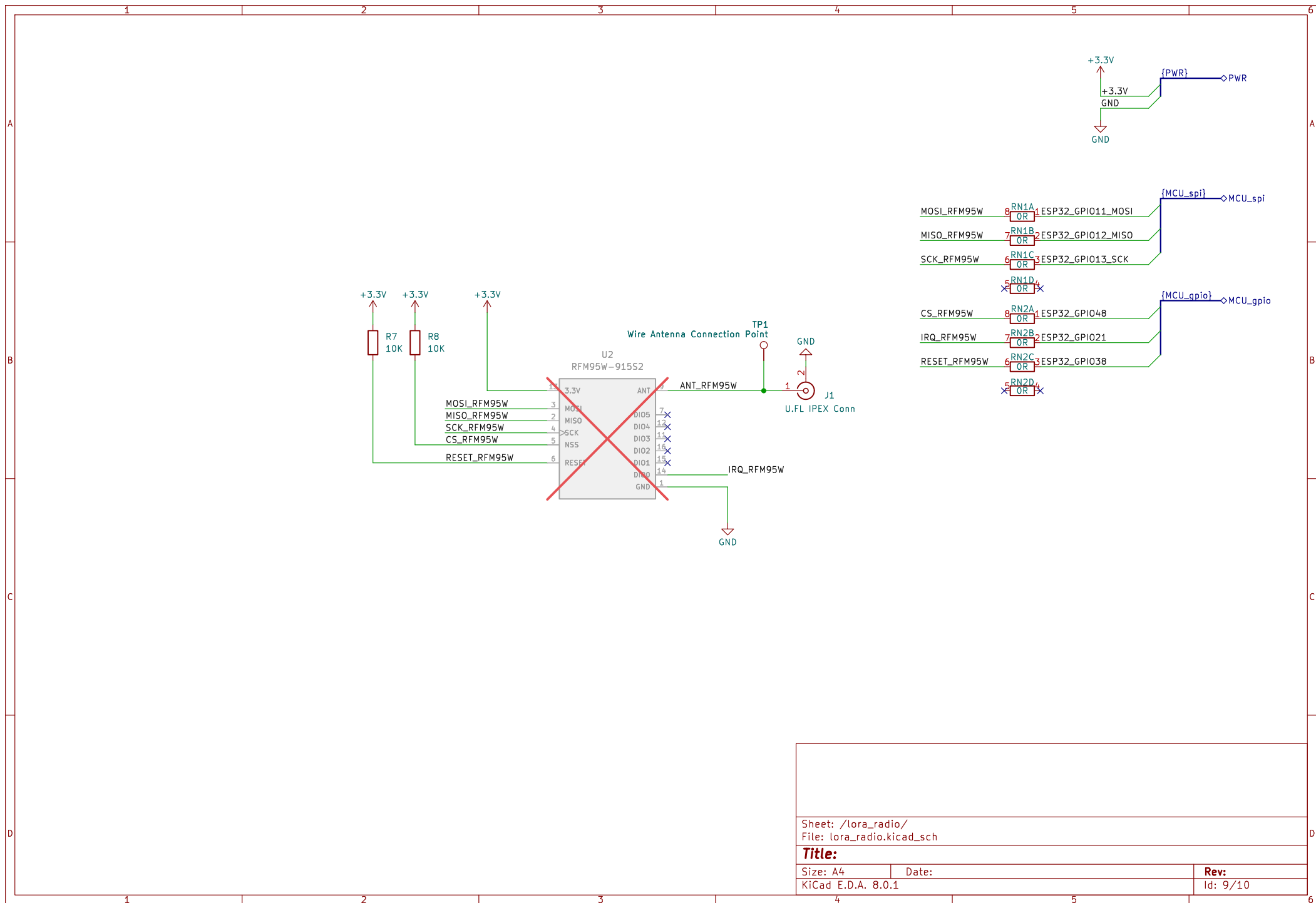
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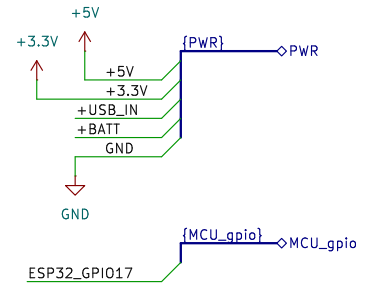
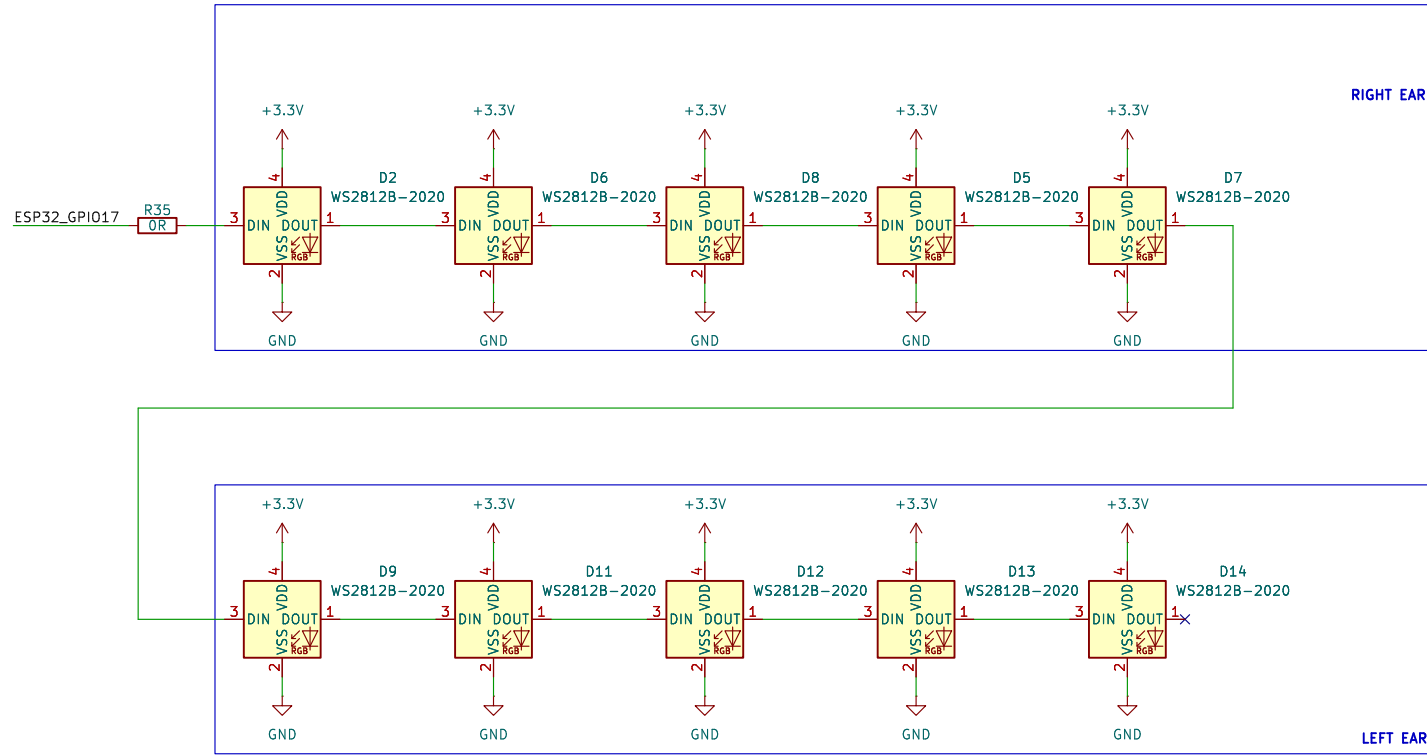
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Size: A4	Date:	Rev:
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